Foundations of Statistical Learning & Algorithms

Today's goal • 3/3 items

Module 1
Introduction to Linear Algebra and Vector Spaces

Getting Started

Course Introduction
Video • 1 min

Course Overview
Reading • 2 min

Syllabus
Reading • 10 min

Meet Your Instructor
Video • 1 min

Meet Your Fellow Learners
Discussion Prompt • 10 min

Academic Integrity
Reading • 1 min

Lesson 1: Introduction to Linear Algebra

Introduction to Machine Learning
Reading • 25 min

Introduction to Linear Algebra
Reading • 4 min

Why Linear Algebra and Mathematics?
Reading • 2 min

Notation
Reading • 5 min

Lesson 2: Systems of Linear Equations

Foundational Concepts of Systems of Linear Equations
Reading • 20 min

[H5P] System of Linear Equations
Ungraded App Item • 20 min

Solved Example and Recommended Resources
Reading • 20 min

Check Your Knowledge: System of Linear Equations
Graded Assignment • Grade: 100%

Lesson 3: Matrices

Matrices and Matrix Operations
Reading • 20 min

Matrices
Video • 8 min

Helpful Resources and Solved Examples
Reading • 10 min

Check Your Knowledge: Matrices
Graded Assignment • 16 min

Lesson 4: Vector Spaces

Foundations of Vector Spaces: Operations and Subspaces
Reading • 20 min

Vector Space
Video • 7 min

Vector Space Properties
Reading • 10 min

Subspace
Reading • 1 min

Check Your Knowledge: Vector Spaces
Graded Assignment • Grade: 85.71%

Lesson 5: Orthogonal Complement and Projection

Introduction to Orthogonal Complement
Reading • 20 min

Introduction to Orthogonal Projections
Reading • 40 min

Importance of Projections
Reading • 4 min

**Projection Onto One-Dimensional Subspaces (Lines)**
Reading • 15 min

Projection Onto General Subspaces
Reading • 10 min

Projections Case Study
Reading • 10 min

Module 2
Linear Transformation and Matrix Decomposition

Module 3
Probability Foundations for Statistical Learning

Module 4
Introduction to Model Evaluation and Optimization

# Projection Onto One-Dimensional Subspaces (Lines)

0:16 / 2:34 2x

**When projecting onto a line through the origin, the projection of a vector $x$ onto a line spanned by $b$ is given by $\pi_U(x) = \frac{\langle x,b\rangle}{\langle b,b\rangle} b$**

.

This projection formula ensures that the error vector $x - \pi_U(x)$ is orthogonal to the subspace, minimizing the distance to the line.

The figure illustrates the projection of a vector $x$ onto a line defined by a vector $b$ in $\mathbb{R}^2$. The diagram shows vector $x$ originating from the origin and vector $b$ defining the line of projection. The projection of $x$ onto the line is labeled $\pi_U(x)$, which lies along the direction of $b$. The vector $x - \pi_U(x)$, representing the component of $x$ orthogonal to $b$, is also depicted, showing that it is perpendicular to the vector $b$ and hence to the line defined by $b$. This figure effectively demonstrates the concept of orthogonal projection, illustrating how $\pi_U(x)$ is the closest point to $x$ on the line, minimizing the distance between $x$ and the subspace spanned by $b$.

**Example :** Find the projection matrix $P_\pi$ onto the line through the origin spanned by

$$\mathbf{b} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

We use the formula for the projection matrix:

$$P_\pi = \frac{\mathbf{b}\mathbf{b}^\top}{\mathbf{b}^\top \mathbf{b}}$$

1. First, compute $\mathbf{b}^\top \mathbf{b}$:

$$b^\top b = 2^2 + 3^2 = 4 + 9 = 13.$$

2. Compute $\mathbf{b}\mathbf{b}^\top$:

$$\mathbf{b}\mathbf{b}^\top = \begin{bmatrix} 2 \\ 3 \end{bmatrix} \begin{bmatrix} 2 & 3 \end{bmatrix} = \begin{bmatrix} 4 & 6 \\ 6 & 9 \end{bmatrix}$$

3. Now calculate $P_\pi$:

$$P_b = \frac{1}{13} \begin{bmatrix} 4 & 6 \\ 6 & 9 \end{bmatrix} = \begin{bmatrix} \frac{4}{13} & \frac{6}{13} \\ \frac{6}{13} & \frac{9}{13} \end{bmatrix}$$

**Projection of a Vector**

Now, let $\mathbf{x} = \begin{bmatrix} 6 \\ 3 \end{bmatrix}$. The projection of $\mathbf{x}$ onto the line spanned by $\mathbf{b}$ is:

$$\pi_U(x) = P_b \mathbf{x}$$

Multiply $P_b$ with $\mathbf{x}$

$$P_b(\mathbf{x}) = \frac{1}{13} \begin{bmatrix} 4 & 6 \\ 6 & 9 \end{bmatrix} \begin{bmatrix} 6 \\ 3 \end{bmatrix}$$

Thus:

$$P_b(\mathbf{x}) = \begin{bmatrix} \frac{42}{13} \\ \frac{63}{13} \end{bmatrix} = \begin{bmatrix} 3.23 \\ 4.85 \end{bmatrix}$$

And the error vector:

$$\mathbf{x} - P_b(\mathbf{x}) \approx [2.77, -1.85]$$

Go to next item Completed

Like Dislike Report an issue

Go to next item